



The impossible diagonal



Class video: <https://youtu.be/4Io4ID81BqE>

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. In Spain the decimal separator is a comma: the board writes 1,414213562 for 1.414213562.

Before watching the video

1. Write each number as a product of prime factors and count how many times 2 appears in each.

Number	Prime factors	How many twos?
12		
25		
50		
144		
288		

25 and 144 are squares ($5 \cdot 5$ and $12 \cdot 12$); 50 and 288 are twice those squares. Do you notice anything their twos have in common?

2. A square has side 1. How long is its diagonal? Why?

3. Look for two whole numbers a and b with $a^2 = 2b^2$. Write down your attempts and how much each one misses by.

a	b	a^2	$2b^2$	$a^2 - 2b^2$

What is your best attempt? How much does it miss by?

While you watch the video

1. What are the decimals of a fraction like? And the other way round?
2. The tablet gives a number almost equal to the square root of 2. Why "almost"? Could it be written exactly, with more digits or with a fraction? Write down what you think now.
3. **Your bet.** If we keep trying, with more digits or with other fractions, will we ever hit it exactly, or will we always miss? Why do you think so?
4. Write in your own words the three steps of a proof by contradiction.
5. If in a square the twos come in pairs, what happens in twice a square? Check it with two numbers of your own.
6. In class 1 you bet on whether, after marking every fraction on the number line, any gap would be left. Was your bet right? Explain in one sentence.

7. "The square root of 2 is irrational because it has infinitely many decimals." What is wrong with that sentence?

8. Why couldn't the tablet's number be exact?

The proof, with gaps

Fill it in without looking at the video.

1 • We assume that $\sqrt{2} = \frac{a}{b}$, with a and b _____ whole numbers.

2 • We reason. Squaring, $a^2/b^2 = 2$. Multiplying both sides by _____, we get $a^2 = 2b^2$. On the left there is a _____: its twos come _____. On the right, _____ a square: it has a two _____.

3 • Contradiction. On both sides of the equals sign there is the _____ number, and a number factorises into primes in _____. It cannot have all its twos in pairs and, at the same time, _____. So the assumption was _____: $\sqrt{2}$ is not a fraction.

Now, in your own words:

1. Why do the twos in a square always come in pairs, whatever the number, and not just with 12?

2. In which step is it used that a number factorises in only one way? What would happen if it were not so?

3. Check it with the first one we tried, $1.4 = 7/5$: $7^2 = \underline{\hspace{2cm}}$ and $2 \cdot 5^2 = \underline{\hspace{2cm}}$. Factorise both numbers. Why can't they be equal?

Practice

1. Prove that the square root of 3 is not a fraction. Hint: instead of twos, count threes.
2. The square root of 4 is a fraction, because it equals 2, which is $2/1$. So if you repeat today's proof with 4, it must fail at some step. Which one?
3. Is $1 + \sqrt{2}$ a fraction? What will its decimals be like?
4. **One more, to revise class 1.** Someone says that $\sqrt{2} - 1 = 0.4142\ 4142\ 4142\dots$, with period 4142. Could they be right?
5. **A challenge.** The tablet's number ends in base 60. Converted to our decimals, $1.41421296296\dots$, it repeats. Why? Hint: it is the fraction $30547/21600$; remember why $1/12$ does not end.

For those who want more: two squares of 12 inside one of 17

On squared paper, draw a square of 17×17 small squares and cut out two of 12×12 . Put one in the bottom left corner and the other in the top right.

1. **Before counting, your prediction.** $12^2 + 12^2 = 288$ and $17^2 = 289$. Will there be more small squares covered twice, or more left uncovered?
2. Count and fill in the first row. Then repeat the move further in: the overlap plays the big square, and the corners play the small ones.

Side of the big square (a)	Side of the small squares (b)	a^2	$2b^2$	Overlap (covered twice)	Corners (uncovered)
17	12				
7	5				
3	2				
1	1				

3. What happens in the last row? What would happen if there were two squares with whole-number sides, one with twice the area of the other?

Keep thinking

The square root of 2 cannot be written out in full with digits, and no fraction catches it. And yet there it is: it is the diagonal of a square. So, are numbers discovered or invented?